MIL-P-13269A(MU)
AMENDMENT 5
10 October 1969
SUPERSEDING
AMENDMENT 4
6 August 1968

MILITARY SPECIFICATION

PERISCOPE, TANK: M20 SERIES

This amendment forms a part of Military Specification MIL-P-13269A (Ord), dated 10 November 1960.

Paragraph 2.1: Under "SPECIFICATIONS" delete "MIL-P-13462 - Periscopes M20 Series; Packaging of" and substitute "MIL-P-14232 - Parts, Equipment and Tools for Ordnance Materiel, Packaging of".

Paragraph 2.1: Under "DRAWINGS" add " and M20A3G" as follows:

"F8585211	Periscopes: M20A3C and M20A3G
LSQAP-8585211	Supplementary Quality Assurance Provisions List for Periscopes: M20A3C and M20A3G
IEL-8585211	Inspection Equipment List for Periscopes: M20A3C and M20A3G"

Paragraph 2.1: Add packaging data sheet after "DRAWINGS" as follows:

"PACKAGING DATA SHEET

MIL-P-14232/P7652791 Packaging of Periscopes, Tank: M20 Series"

Add the following paragraph after paragraph 3.6.5:

"3.6.6 Torque.- The running torque required to rotate the coupling through maximum elevation and depression shall not exceed 3 pound inches at a standard ambient temperature of 60° to 90°F. The torque shall not change from the value obtained by more than 2 pound inches when subjected to the temperature specified in 3.5.2."

* Paragraph 3.7: Delete in its entirety and substitute:

"3.7 Body assembly."

* Paragraph 3.7.1: Delete and substitute

"3.7.1 Boresight Travel, except M20A3C and M20A3G.- With a target line of sight perpendicular to datum line "B" and parallel to mounting surface "A", the line of sight through the boresight mark shall coincide with the target image within 1. mil, from this reticle setting. The elevation and deflection boresight knobs shall provide a minimum of 7. mils travel in each direction from the position of target and reticle coincidence. The elevation and deflection boresight knob scales shall be set to read "3"."

FSC 1240

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* Paragraph 3.7.1.1: Delete and substitute

"3.7.1.1 Boresight Travel, M20A3C and M20A3G only.- With a target line of sight perpendicular to datum line "B" and making an angle of 35.2 mils in elevation relative to datum plane "A" (measured counter-clockwise from datum plane "A" with eyepiece on the right side), the line of sight through the boresight mark shall coincide with the target image within 1. mil. From this reticle setting, the elevation and deflection boresight knobs shall provide a minimum of 7. mils travel in each direction from the position of target and reticle coincidence. The elevation and deflection boresight knob scales shall be set to read "3"."

Add the following paragraph:

"3.7.1.2 Boresight locks.- Actuation of the locks of the elevation and deflection boresight knobs shall not cause more than 0.05 mil shift in the line of sight and not more than 0.05 mil rotation of the boresight knob, as measured on the knob scale."

Add the following paragraph:

"3.9 Sealing.- The periscope (head assembly and body assembly) shall show no evidence of leakage when subjected to an external pressure of 5.0 ± 0.1 pounds per square inch gage (psig)."

Paragraphs 4.4 and 4.4.1: Delete and substitute the following:

"4.4 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) periscopes, plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.4.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.4.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

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Paragraph 4.7.2: Delete "Table - Classification of Defects" and substitute "Table I - Classification of Defects".

* Paragraph 4.7.2, TABLE II:

Add the following characteristic

"107a. Torque 3.6.6 4.9.2.6"

Delete Characteristic "108" and substitute

"108. Boresight travel, 3.7.1 4.9.3.1" except M20A3C and M20A3G

"108a. Boresight travel, 3.7.1.1 4.9.3.1.1" M20A3C and M20A3G only

"108b. Boresight locks 3.7.1.2 4.9.3.1.2"

* Paragraph 4.8.3.4, paragraph heading: Delete "collimation" and substitute "boresight travel"

Add the following paragraph

"4.9.2.6 Torque.- The torque required to rotate the input coupling through the specified excursion shall be measured using standard measuring equipment affixed to the coupling. The measured torque shall not exceed the value specified in 3.6.6 for the temperature indicated in 3.5.2."

* Paragraph 4.9.3: Delete and substitute

"4.9.3 Body assembly.- The following tests shall be performed with the body assembly positioned as specified in 3.7.1 or 3.7.1.1 on a special testing device conforming to 4.8.3.4 for the applicable periscope under test. Unless otherwise specified, the elevation and deflection boresight knobs shall be in the unlocked position."

* Paragraph 4.9.3.1: Delete and substitute

"4.9.3.1 Boresight travel, except M20A3C and M30A3G.- With backlash removed for direction of travel, aline the 3 mil graduations of the knob scales to their fixed indexes without overtravel. While viewing through the eyepiece of the body assembly, obtain best focus of the target image by adjustment of the diopter motion scale, observe the line of sight through the boresight mark of the body assembly reticle to the target image. The boresight mark (intersecting point of the horizontal and verticle reticle lines) shall be in coincidence with, or within, the lines of the collimating square on the target image to determine conformance with the applicable requirement of 3.7.1. From this established position of the reticle setting, move the line of sight by rotating each respective boresight knob to its clockwise stop, then to its counterclockwise stop using the knob scales to record the mil travel of the boresight mark. These rotations of the elevation and deflection boresight knobs shall move the reticle boresight mark the distance specified in 3.7.1, as measured on the boresight knob scales. The elevation and deflection boresight knobs may be in the locked position while slipping the boresight knob scales."

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* Paragraph 4.9.3.1.1: Delete and substitute

"4.9.3.1.1 Boresight travel, M20A3C and M20A3G only.- Perform this test in accordance with the procedure specified in 4.9.3.1. The requirements of 3.7.1.1 shall be met."

Add the following paragraph:

"4.9.3.1.2 Boresight locks.- Adjust the elevation boresight knob so that a line of sight through the boresight mark of the body assembly reticle is coincident with a selected point on the horizontal line of the target image collimating square. Record reading of elevation boresight knob. Engage elevation boresight knob lock. Again record reading of elevation boresight knob. Any difference between the first and second recorded reading on the elevation boresight knob, is the amount of knob rotation in elevation. Any boresight knob rotation shall not exceed the limit specified in 3.7.1.2. Observe the line of sight through the boresight mark of the body assembly reticle to the horizontal line of the target image and observe for any lack of coincidence. Any shift in the line of sight through the boresight mark shall not exceed the limit specified in 3.7.1.2 as measured on the target. Repeat this test with the deflection boresight knob using the vertical line of the target image collimating square and the associated knob scale to record the readings. Any shift in the line of sight, or rotation of the deflection boresight knob, when the deflection boresight knob lock is engaged, shall not exceed the applicable tolerance specified in 3.7.1.2."

Paragraph 4.9.5: Delete and substitute

"4.9.5 Sealing test.- The external pressurizing of the periscope head and body assemblies to test sealing shall be performed with special testing equipment conforming to Drawings F7680673, F7680674 and F7680675. The test shall be conducted in accordance with the instructions on Sheet 1 of Drawing F7680673. The assemblies shall be subjected to the specified pressure for a period of 5 minutes ± 15 seconds during which time there shall be no evidence of leakage as indicated by the testing equipment. Sampling for examination and tests shall be in accordance with MIL-STD-105 utilizing an AQL of 0.65 percent in conjunction with Table II-A of MIL-STD-105.

Add the following paragraph:

4.9.6 Packaging, packing, and marking.- Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein."

Paragraph 5.1: Delete "Specification MIL-P-13462" and substitute "MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P7652791."

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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